

Problem

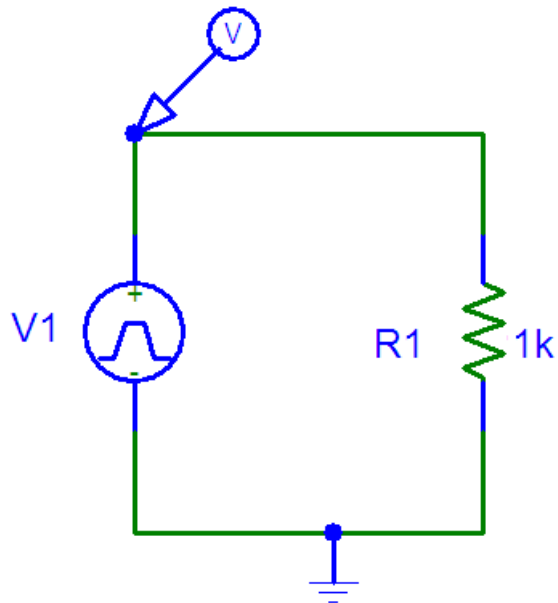
Use *PSpice* or *MultiSim* to find the Fourier components of the signal in Prob. 17.7.

Step-by-step solution

Step 1 of 7

Using PSpice to find the Fourier components consists of four steps.

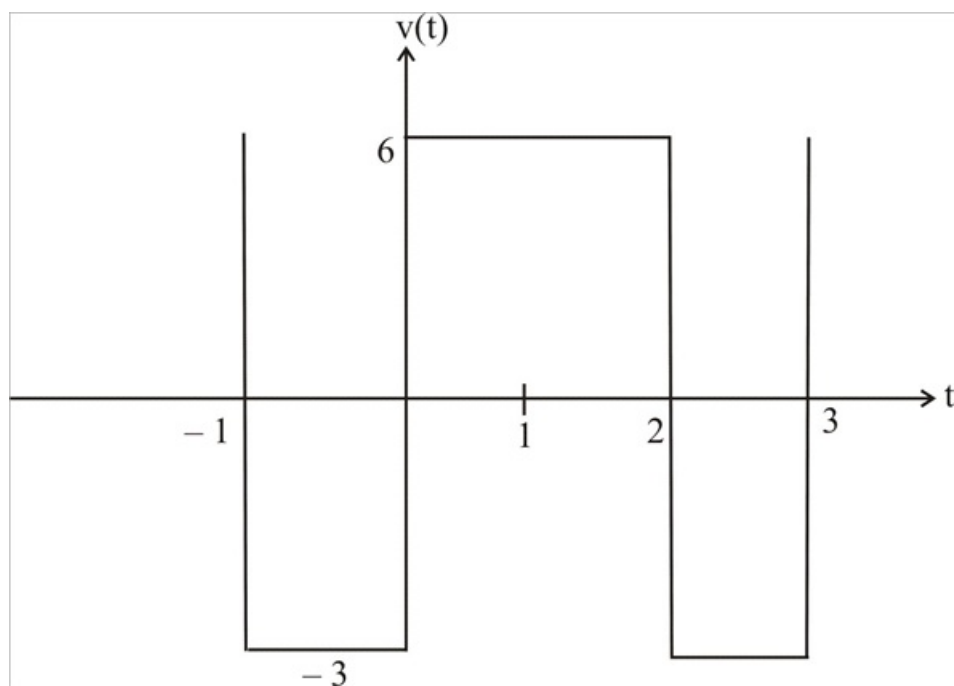
Step1:-



The schematic for obtaining the fourier coefficients

Step 2 of 7

The signal given is,



Step 3 of 7

Step2:- We enter the attributes of the voltage source VPULSE by double clicking on the voltage source and enter the values as

$$V1 = -3$$

$$V2 = 6$$

$$TD = 0$$

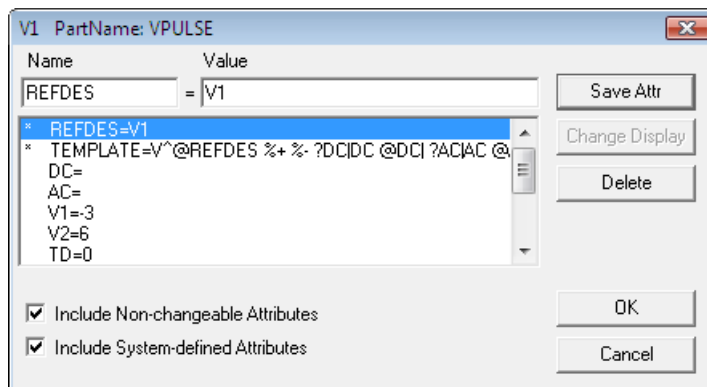
$$TR = 1\mu s$$

$$TF = 1\mu s$$

$$PW = 2$$

$$PER = 3$$

Step 4 of 7



Step 5 of 7

Step3:- The given signal is evident that $T = 3s$

$$f_0 = \frac{1}{T}$$

$$f_0 = \frac{1}{3}$$

$$f_0 = 0.33Hz$$

In the menu bar

click **analysis** → **setup**

In the setup disable the Bias point. And enable the **transient** and click on transient and enter the parameter as

Final time as $4T = 12s$

Print step as $0.01s$

Step ceiling as $10m$

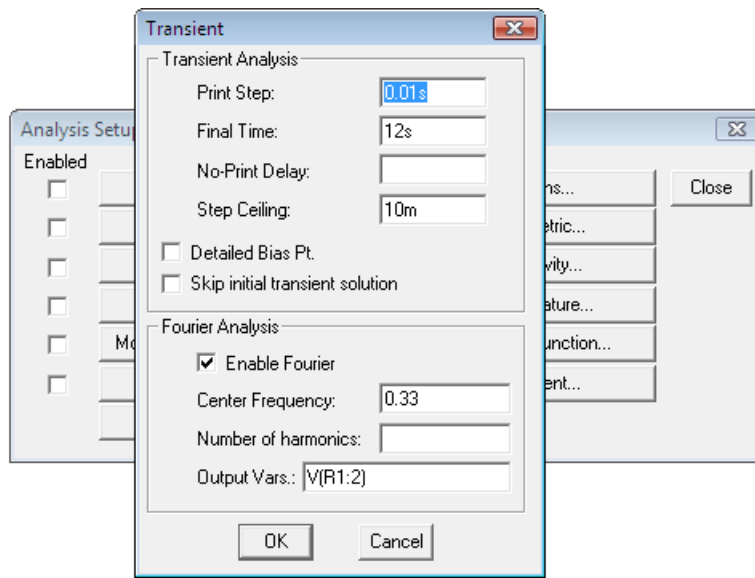
In the **Fourier analysis**

Tick **Enable Fourier**

Centre frequency as 0.33 and the

Output VARS as $V(R1:2)$

Step 6 of 7



Step 7 of 7

Step4:- Then same and click on simulate, when PSpice is run, output file contains the result as

FOURIER COMPONENTS OF TRANSIENT RESPONSE V(\$N_0001)

DC COMPONENT = 2.940594E+00

HARMONIC FREQUENCY FOURIER NORMALIZED PHASE NORMALIZED

NO (HZ) COMPONENT COMPONENT (DEG) PHASE (DEG)

1 3.300E-01 5.020E+00 1.000E+00 -3.297E+01 0.000E+00

2 6.600E-01 2.420E+00 4.820E-01 2.406E+01 9.000E+01

3 9.900E-01 1.188E-01 2.365E-02 -9.891E+01 3.937E-07

4 1.320E+00 1.296E+00 2.581E-01 -4.188E+01 9.000E+01

5 1.650E+00 9.282E-01 1.849E-01 1.515E+01 1.800E+02

6 1.980E+00 1.186E-01 2.362E-02 -1.078E+02 9.000E+01

7 2.310E+00 7.613E-01 1.516E-01 -5.079E+01 1.800E+02

8 2.640E+00 5.532E-01 1.102E-01 6.238E+00 2.700E+02

9 2.970E+00 1.183E-01 2.356E-02 -1.167E+02 1.800E+02